



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

Semester II 2023/2024

Subject : PROBABILITY & STATISTICAL DATA ANALYSIS (SECI 1143)
Task : Chapter 7 (Correlation and Regression) - **85 marks**
Due Date : **1 week after release (please refer to the section's lecturer)**

INSTRUCTION:

1. This is a **GROUP** assignment. Please clearly write your member's **NAME & MATRIC NUMBER** on the front page of the submission.
2. This assignment contributes to **5%** of overall course marks.
3. Only **HANDWRITTEN** submission is accepted:
 - a. Submissions using any reporting or statistical tools (e.g.: MS Word, MS Excel, etc.,) will be **REJECTED**.
 - b. Make sure the submission is neatly written. Any submission with handwriting that is unreadable, will be **REJECTED**.
 - c. For answers that need to draw graphs, using graph paper is optional. You can use plain paper.
 - d. Round your answers to **THREE** decimal places.
4. Submission **online** or in **hard copy** (please refer to the section's lecturer).

QUESTION 1

(15 MARKS)

A biscuit production manager wants to investigate the linear relationship between the number of items produced and the production cost. To pursue his/her objective, the manager recorded the data on the number of items produced per day and the production cost per day (in thousands of Malaysian Ringgit).

Table 1 shows the data collected for 12 consecutive days.

Days	1	2	3	4	5	6	7	8	9	10	11	12
Number of items, (x)	26	44	53	29	77	80	20	40	67	86	17	61
Production cost, (y)	42	60	69	47	91	98	39	55	85	104	37	77

- a. Name the **TWO (2)** most popular correlation coefficient. (2 marks)
- b. Calculate the correlation coefficient, r . (11 marks)

- c. Based on the correlation coefficient, r obtained, make a conclusion on the linear relationship between the number of items produced and the production cost

(2 marks)

$$r = \frac{\sum xy - (\sum x \sum y)/n}{\sqrt{[(\sum x^2) - (\sum x)^2/n][(\sum y^2) - (\sum y)^2/n]}}$$

QUESTION 2

(10 MARKS)

A local authority is going to study the relationship between the size of households and the daily plastic usage consumed by them. The results of a sampling given below in Table 5 indicate some values of X (weight of plastic usage) and Y (the size of household).

Table 2: Weight of plastic usage (X) and size of household (Y)

X	0.27	1.41	2.19	2.83	2.19	1.81	0.85	3.05
Y	2	3	3	6	4	2	1	5

- From the above sampling, find the value of correlation coefficient r . (5 marks)
- Using the same sample data set above, conduct the hypothesis testing to know whether the variables X and Y correlate using a 95% confidence level. (3 marks)
- Find out if the decision in (ii) may change if you increase the confidence level to 99%. (2 marks)

QUESTION 3

(25 MARKS)

A social media analytics company is investigating the relationships between various factors to understand how they influence the engagement on posts. The company has collected data on the following variables (Table 3) for 6 different social media posts:

- Number of Likes:** The number of likes received by the post.
- Number of Comments:** The number of comments received by the post.
- Number of Shares:** The number of times the post was shared.
- Post Length:** The length of the post in characters.
- Engagement Score:** A score (from 1 to 100) representing the overall engagement on the post.
- Sentiment Score:** An ordinal variable representing the sentiment of the post (1: Very Negative, 2: Negative, 3: Neutral, 4: Positive, 5: Very Positive).

Table 3: Factors influencing engagement on posts

Post ID	Likes	Comments	Shares	Post Length	Engagement Score	Sentiment Score
1	150	20	30	200	85	4
2	100	10	20	100	70	3
3	200	25	40	250	90	5
4	80	8	15	150	60	2
5	170	22	35	220	88	5
6	120	15	25	180	75	3

- Name two variables whose relationship can be expressed by Spearman's Rho Rank correlation coefficient. (2 marks)
- Name two variables whose relationship can be expressed by the Pearson correlation coefficient. (2 marks)
- Compute the Spearman correlation coefficients for the same variables in (a). (8 marks)
- Identify and compare the strongest positive and negative correlations from:
 - Likes and comments
 - Likes and share
 - Engagement Score and Sentiment Score (6 marks)
- Test the null hypothesis that there is no correlation between 'Engagement Score' and 'Post Length' against the alternative hypothesis that there is a significant correlation. Use a significance level of 0.05 and provide the test statistics and p-values. (7 marks)

QUESTION 4**(15 MARKS)**

Last year, five randomly selected students took a math aptitude test before they began their statistics course. The Statistics Department would like to analyze the relationship of the data, make predictions using the regression equation and validate the regression equation. The data of the math aptitude test score and corresponding statistics grade are as shown in Table 4 below:

Table 4: Score on Math and Statistics grade

Score on math aptitude test (x)	Statistics grade (y)
95	85
85	95
80	70
70	75
65	70

Answers at 2 decimal places.

- Calculate $\sum x$, $\sum y$, $\sum xy$ and $\sum x^2$ (4 marks)
- Calculate value of b_1 and b_0 (2 marks)
- Based on answers in (a) and (b), what linear regression equation best predicts statistics performance, based on math aptitude scores? (2 marks)
- Which graph represents the regression equation in (c)? (1 mark)
- If a student made a 60 on the aptitude test, what grade would we expect her to make in statistics? (2 marks)
- Find the value of SSR, SST and R^2 . (3 marks)
- Based on your answer in (f), how well does the regression equation fit the data? (1 mark)

QUESTION 5

(20 MARKS)

A software development company wants to predict the time required to complete new software projects based on the estimated lines of code (LOC). The company has collected data on the development time (in weeks) and LOC for previous projects. Using this data presented in Table 5, answer the following questions in 2 decimal places:

Table 5: Length of code and development time

Project ID	LOC (K)	Development Time (weeks)
1	150	40
2	100	35
3	200	50
4	80	30
5	170	45
6	120	38
7	160	42
8	90	28
9	250	55
10	130	37

- Plot a scatter plot of the data with LOC on the x-axis and Development Time on the y-axis. (3 marks)
- Calculate the correlation coefficient between LOC and Development Time. (7 marks)
- Fit a simple linear regression model using LOC as the independent variable and Development Time as the dependent variable. Provide the regression equation and interpret the coefficients. (5 marks)
- Calculate the R-squared value and interpret it. (3 marks)
- Use the regression model to predict the development time for a new project with an estimated 180K LOC. (2 marks)

1. a) Pearson's product - moment correlation coefficient
Spearman's rho rank correlation coefficient

b)	Num of items (x)	Production cost (y)	xy	x^2	y^2
	26	42	1092	676	1764
	44	60	2640	1936	3600
	53	69	3657	2809	4761
	29	47	1363	841	2209
	77	91	7007	5929	8281
	80	98	7840	6400	9604
	20	39	780	400	1521
	40	55	2200	1600	3025
	67	85	5695	4489	7225
	86	104	8944	7396	10816
	17	37	629	289	1369
	61	77	4697	3721	5929
	$\sum x = 600$	$\sum y = 804$	$\sum xy = 46544$	$\sum x^2 = 36486$	$\sum y^2 = 60104$

$$\begin{aligned}
 r &= \frac{\sum xy - (\sum x \sum y)/n}{\sqrt{(\sum x^2 - (\sum x)^2/n)(\sum y^2 - (\sum y)^2/n)}} \\
 &= \frac{46544 - [(600)(804)]/12}{\sqrt{(36486 - (600)^2/12)(60104 - (804)^2/12)}} \\
 &= \frac{6344}{\sqrt{40446696}} \\
 &= 0.998
 \end{aligned}$$

- c) Since $r = 0.998$, there is a relatively strong positive linear relationship between the number of items produced and the production cost.

2. a)

X	Y	XY	X ²	Y ²
0.27	2	0.54	0.073	4
1.41	3	4.23	1.988	9
2.19	3	6.57	4.796	9
2.83	6	16.98	8.009	36
2.19	4	8.76	4.796	16
1.81	2	3.62	3.276	4
0.85	1	0.85	0.723	1
3.05	5	15.25	9.303	25
$\sum x = 14.6$	$\sum y = 26$	$\sum xy = 56.8$	$\sum x^2 = 32.964$	$\sum y^2 = 104$

$$r = \frac{56.8 - (14.6)(26)/8}{\sqrt{(32.964 - (14.6)^2/8)(104 - (26)^2/8)}}$$

$$= \frac{9.35}{\sqrt{123.2205}}$$

$$= 0.842$$

b) $H_0: \rho = 0$ (no linear correlation)
 $H_1: \rho \neq 0$ (linear correlation exists)

$$t = \frac{0.84}{\sqrt{\frac{1 - 0.842^2}{8 - 2}}}$$

$$= 3.814$$

$$\alpha = 0.05, df = 8 - 2 = 6$$

$$t_{0.025, 6} = 2.447$$

$\therefore$ Since $3.814 > 2.447$, we reject H_0 . There is sufficient evidence at 0.05 significance level to that there is linear relationship between the weight of plastic usage and size of household.

c) $t_{0.005, 6} = 3.707$

$\therefore$ Since $3.814 > 3.707$, we reject H_0 . The decision in (ii) is still same.

3.a) Engagement score and Sentiment score

b) Number of likes and number of comments

c)	Post ID	Rank of Engagement score	Rank of Sentiment score	d_i	d_i^2
	1	3	3	0	0
	2	5	4.5	0.5	0.25
	3	1	1.5	-0.5	0.25
	4	6	6	0	0
	5	2	1.5	0.5	0.25
	6	4	4.5	-0.5	0.25

$$\sum d_i^2 = 1$$

$$r_s = 1 - \frac{6(1)}{6(6^2 - 1)}$$

$$= 0.971$$

d) a)	likes (x)	Comments (y)	xy	x^2	y^2
	150	20	3000	22500	400
	100	10	1000	10000	100
	200	25	5000	40000	625
	80	8	640	6400	64
	170	22	3740	28900	484
	120	15	1800	14400	225
	$\sum x = 820$	$\sum y = 100$	$\sum xy = 15180$	$\sum x^2 = 122200$	$\sum y^2 = 1898$

$$r = \frac{15180 - (820)(100)/6}{\sqrt{(122200 - (820)^2/6)(1898 - (100)^2/6)}}$$

$$= \frac{1513.33}{\sqrt{2344177.778}}$$

$$= 0.988$$

b)	Likes (x)	Share (y)	xy	x^2	y^2
	150	30	4500	22500	900
	100	20	2000	10000	400
	200	40	8000	40000	1600
	80	15	1200	6400	225
	170	35	5950	28900	1225
	120	25	3000	14400	625
	$\sum x = 820$	$\sum y = 165$	$\sum xy = 24650$	$\sum x^2 = 122200$	$\sum y^2 = 4975$

$$r = \frac{24650 - (820)(165)/6}{\sqrt{(122200 - (820)^2/6)(4975 - (165)^2/6)}}$$

$$= \frac{2100}{\sqrt{443333.33}}$$

$$= 0.997$$

$$c) r = 1 - \frac{6(1)}{6(6^2-1)}$$

$$= 0.971$$

∴ The strongest positive correlations is between likes and shares, which $r = 0.997$ that is closer to 1.

Likes and comments, $r = 0.988$ > Engagement score and Sentiment score, $r = 0.971$

There are no negative correlations as all r are positive.

e) $H_0: \rho = 0$ (No linear correlation)

$H_1: \rho \neq 0$ (Linear correlation exists)

Engagement score (x)	Post length (y)	xy	x^2	y^2
85	200	17000	7225	40000
70	100	7000	4900	10000
90	250	22500	8100	62500
60	150	9000	3600	22500
88	220	19360	7744	48400
75	180	13500	5625	32400
$\sum x = 468$	$\sum y = 1100$	$\sum xy = 88360$	$\sum x^2 = 37194$	$\sum y^2 = 216800$

$$r = \frac{88360 - (468)(1100)/6}{\sqrt{(37194 - (468)^2/6)(216800 - (1100)^2/6)}}$$

$$= \frac{2560}{\sqrt{9752000}}$$

$$= 0.8197$$

$$t = \frac{0.8197}{\sqrt{\frac{1 - (0.8197)^2}{6-2}}}$$

$$= 2.862$$

$$\alpha = 0.05, df = 6 - 2 = 4$$

$$t_{0.025, 4} = 2.776$$

∴ Since $2.862 > 2.776$, we reject H_0 . There is sufficient evidence at 0.05 significance level to support that there is correlation between 'Engagement Score' and 'Post length'.

4.

x	y	xy	x ²	y
95	85	8075	9025	88.76
85	95	8075	7225	82.66
80	70	5600	6400	79.61
70	75	5250	4900	73.51
65	70	4550	4225	70.46
$\Sigma x = 395$	$\Sigma y = 395$	$\Sigma xy = 31550$	$\Sigma x^2 = 31775$	

$$\bar{x} = 79 \quad \bar{y} = 79$$

a) $\Sigma x = 395$

$\Sigma y = 395$

$\Sigma xy = 31550$

$\Sigma x^2 = 31775$

c) $\hat{y} = 30.81 + 0.61x$

d) Scatter plot and regression line with intercept of 30.81 and slope of 0.61

$$b) b_1 = \frac{\Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n}}{\Sigma x^2 - \frac{(\Sigma x)^2}{n}}$$

$$e) \hat{y} = 30.81 + 0.61(60) = 67.41$$

$b_1 = 0.61$

$b_0 = \bar{y} - b_1 \bar{x}$

$= 79 - 0.61(79)$

$= 30.81$

$$f) SSR = \Sigma (\hat{y}_i - \bar{y})^2$$

$$= (88.76 - 79)^2 + (82.66 - 79)^2 + (79.61 - 79)^2 + (73.51 - 79)^2 + (70.46 - 79)^2$$

$$= 212.10$$

$$SST = \Sigma (y_i - \bar{y})^2$$

$$= (85 - 79)^2 + (95 - 79)^2 + (70 - 79)^2 + (75 - 79)^2 + (70 - 79)^2$$

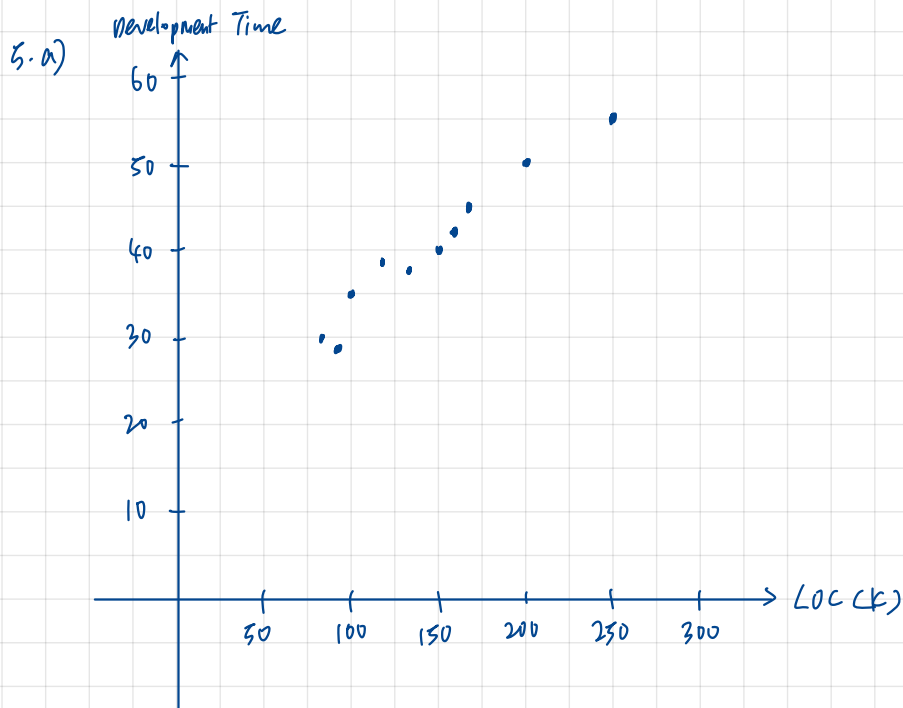
$$= 470$$

$$R^2 = \frac{SSR}{SST}$$

$$= \frac{212.10}{470}$$

$$= 0.45$$

g) R^2 value of 0.45 indicates that 45% of variation in statistic grade is explained by math aptitude test scores. It suggest a moderate fit of regression equation to the data.



LOC (x)	Development Time (y)	xy	x ²	y ²
150	40	6000	22500	1600
100	35	3500	10000	1225
200	50	10000	40000	2500
80	30	2400	6400	900
170	45	7650	28900	2025
120	38	4560	14400	1444
160	42	6720	25600	1764
90	28	2520	8100	784
250	55	13750	62500	3025
130	37	4810	16900	1369
$\Sigma x = 1450$	$\Sigma y = 400$	$\Sigma xy = 61910$	$\Sigma x^2 = 235300$	$\Sigma y^2 = 16636$

$$b) r = \frac{61910 - (1450)(400)/10}{\sqrt{(235300 - 1450^2/10)(16636 - 400^2/10)}}$$

$$= \frac{3910}{\sqrt{15931800}}$$

$$= 0.98$$

$$d) R^2 = r^2$$

$$= (0.98)^2$$

$$= 0.96$$

$$c) b_1 = \frac{61910 - (1450)(400)/10}{235300 - 1450^2/10}$$

$$= 0.16$$

$$\bar{y} = \frac{400}{10} = 40$$

$$\bar{x} = \frac{1450}{10} = 145$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$= 40 - 0.16(145)$$

$$= 16.8$$

$$\therefore \hat{y} = 16.8 + 0.16x$$

$$e) \hat{y} = 16.8 + 0.16(180)$$

$$= 45.6 \text{ weeks}$$